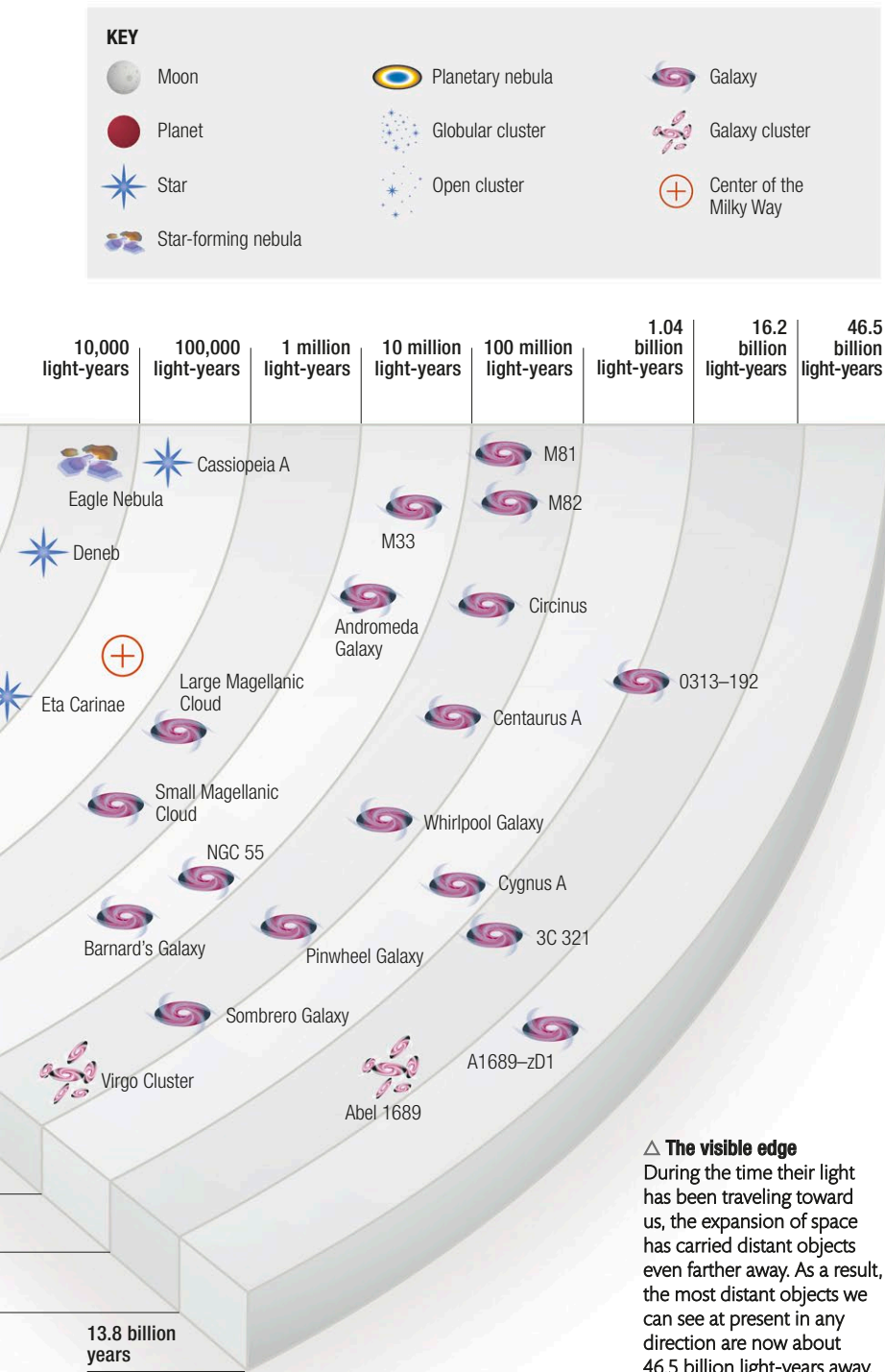
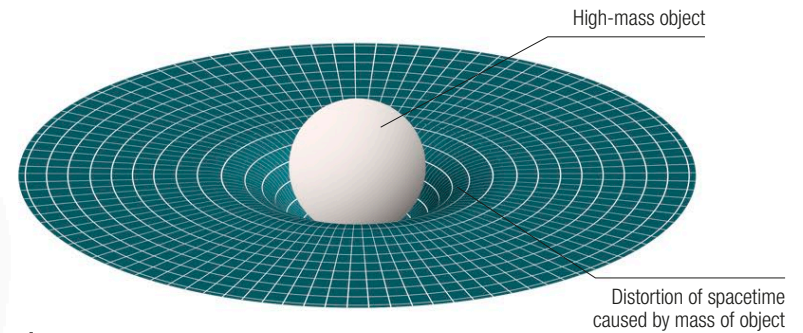




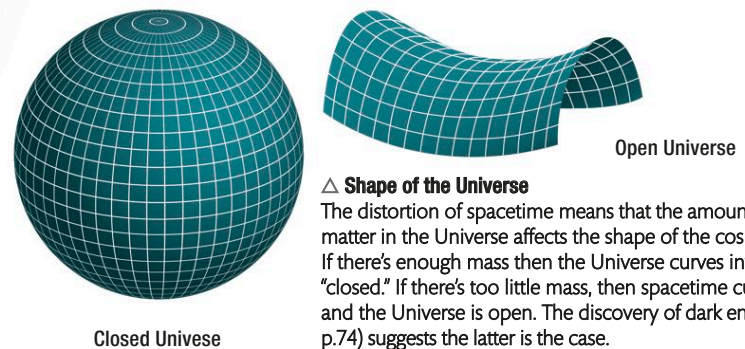
Spacetime

Albert Einstein's theories of special and general relativity explain the fabric of the Universe as a four-dimensional "manifold," in which three familiar dimensions of space (length, breadth, and height) can be traded off with one another and with the fourth dimension of time itself. This is only apparent in extreme situations, such as when objects move at close to the speed of light, but also when large amounts of mass are present. According to Einstein, the force of gravity around massive objects is a result of the way they distort spacetime.



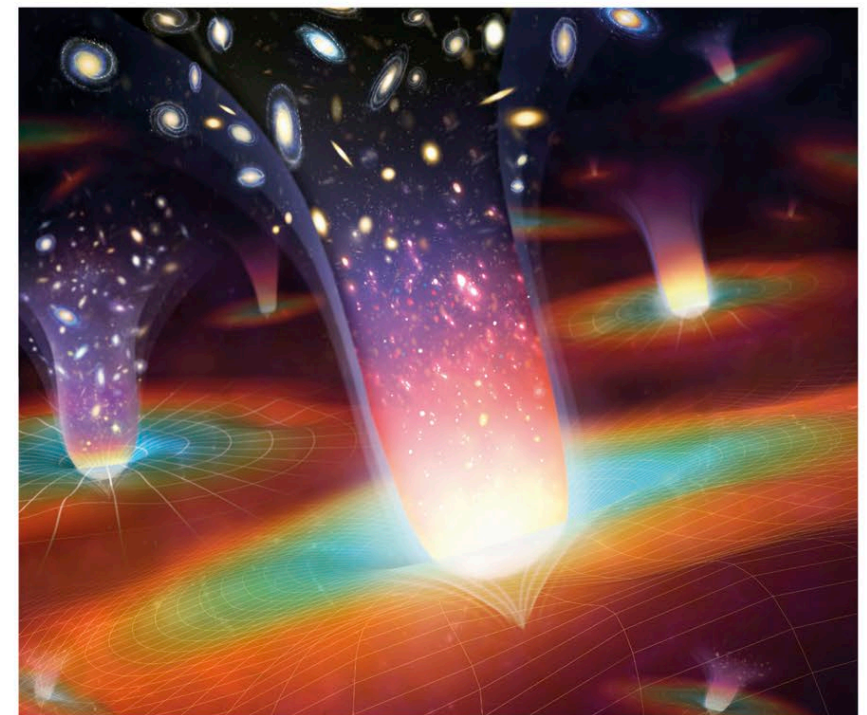
△ Warped space

The idea of spacetime is hard to visualize in four dimensions, but it gets easier if you visualize space as a flat rubber sheet. Massive objects create dents in the sheet (gravitational fields), and these deflect the paths of others moving nearby, or even rays of light (see p.64).



△ Shape of the Universe

The distortion of spacetime means that the amount of matter in the Universe affects the shape of the cosmos itself. If there's enough mass then the Universe curves inward and is "closed." If there's too little mass, then spacetime curves out and the Universe is open. The discovery of dark energy (see p.74) suggests the latter is the case.



More than one universe

Our familiar cosmos certainly stretches beyond the limits of what we can see, but is this Universe the only one? Or, are we part of a wider multiverse? One theory, known as eternal inflation, suggests that our Universe is just one of many. Bursts of inflation energy, such as the one at the beginning of our Universe, are continuously producing new "bubble Universes."

▷ Eternal inflation?

If our Universe is one of many created from the same raw material, then it's possible that the walls of separate bubble Universes would occasionally collide and interact.